

How much of a cross-sectional equity signal survives honest validation, and does it carry to ETFs and index futures?

Question. The algo-wheel project built an ordinary machine-learning signal on 156 US large caps (eight price-and-volume features, ridge and gradient boosting, refit each January) to study what execution costs do to it. Here the signal itself is the object: how much of its information coefficient is real once validation is honest, which features carry it, how fast it decays, what a risk model leaves of it, what the algo-wheel's own cost model leaves of it at a stated capital, and whether the same feature set, fitted on equities, says anything about ETFs and futures.

Answer. (1) Out of sample over 2010-2026 the 5-day IC is 0.0177 with a block-bootstrap band [0.0058, 0.0299] and a Newey-West t of 3.1 over 4,197 days; the in-sample IC of the training windows runs 0.041. The same learner reports 0.023 under shuffled K-fold, 0.0220 under purged and embargoed K-fold and 0.0177 walk-forward: the leak is the difference between the first and the second. (2) The signal's rank autocorrelation is 0.48 at five days (half-life 5 days), and its IC against a single day's return falls from 0.0081 one day ahead to 0.0002 ten days ahead. (3) Neutralising beta, size and sector leaves 0.0086; neutralising momentum and volatility as well leaves 0.0085, and the gross rank book has a factor R-squared of 0.56 with an alpha t of 2.1. (4) At \$1bn the rank book earns 6.0 % a year gross (Sharpe 0.74) and 2.7 % net of the routed cost model (Sharpe 0.33) at 47 % turnover per rebalance; the net IC is 0.0009 [-0.0111, 0.0130]; breakeven capital \$3.6bn; the deflated Sharpe over 41 trials is 0.16. (5) The equity-fitted model applied unchanged gives an IC of 0.0218 on 92 ETFs and 0.0238 on 34 futures; refitted within each universe, 0.0256 and 0.0098.

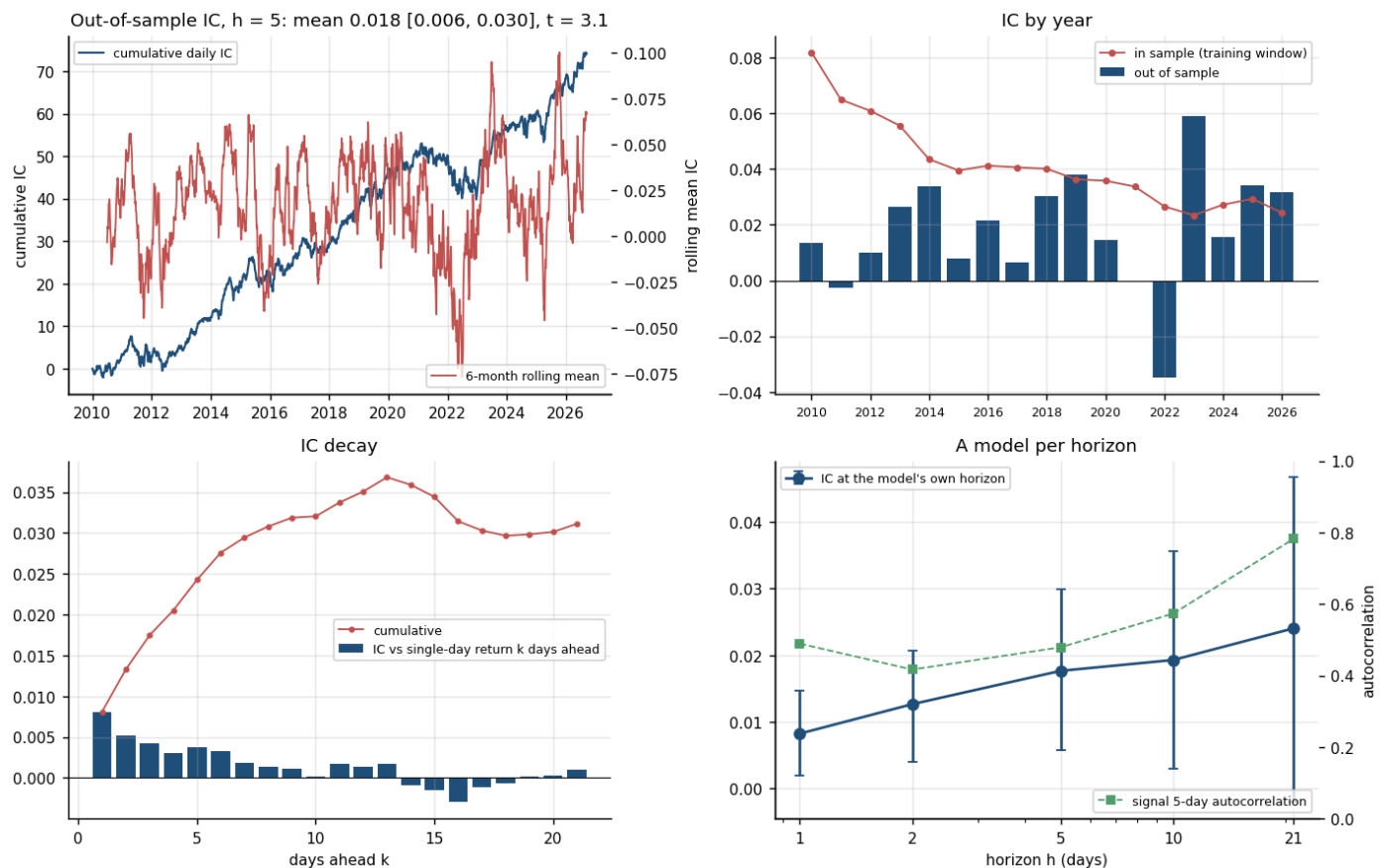


Figure 1. The signal out of sample: cumulative and rolling IC; IC by year against the in-sample IC of each training window; IC decay by days ahead; a model per horizon with the signal's five-day autocorrelation.

Validation ladder and ablations

blend, h = 5	ic_mean	ic_sd	n_days
in_sample	0.0264	0.2509	4197.0000
kfold_shuffled	0.0232	0.2428	4197.0000
kfold_blocked	0.0201	0.2469	4197.0000
purged	0.0220	0.2426	4197.0000
walk_forward	0.0177	0.2097	4197.0000

blend, h = 21	ic_mean	ic_sd	n_days
in_sample	0.0465	0.2460	4181.0000
kfold_shuffled	0.0438	0.2398	4181.0000
kfold_blocked	0.0359	0.2402	4181.0000
purged	0.0359	0.2401	4181.0000
walk_forward	0.0241	0.2111	4181.0000

flexible gbm, h = 5	ic_mean	ic_sd	n_days
in_sample	0.0395	0.2196	4197.0000
kfold_shuffled	0.0248	0.2012	4197.0000
kfold_blocked	0.0155	0.2091	4197.0000
purged	0.0168	0.1918	4197.0000
walk_forward	0.0120	0.1632	4197.0000

flexible gbm, h = 21	ic_mean	ic_sd	n_days
in_sample	0.1108	0.1985	4181.0000
kfold_shuffled	0.0737	0.1925	4181.0000
kfold_blocked	0.0204	0.1905	4181.0000
purged	0.0240	0.1984	4181.0000
walk_forward	0.0133	0.1486	4181.0000

variant	ic_mean	t_nw	delta_vs_full	delta_t
full	0.0177	3.0697	nan	nan
drop_mom_12_1	0.0158	2.7002	-0.0019	-2.0713
drop_rev_1m	0.0176	3.0290	-0.0001	-0.2481
drop_rev_1w	0.0163	2.5711	-0.0014	-0.5947
drop_vol_20	0.0170	2.9093	-0.0007	-0.9198
drop_vol_ratio	0.0182	3.1456	0.0005	0.8112
drop_volume_trend	0.0181	3.1037	0.0004	0.9933
drop_range_20	0.0182	3.1814	0.0006	0.6237
drop_hi_52w	0.0167	2.9381	-0.0010	-0.6636
only_momentum	0.0171	2.7395	-0.0006	-0.1934
drop_momentum	0.0181	3.1852	0.0004	0.2626
only_reversal	0.0156	3.3536	-0.0021	-0.4403
drop_reversal	0.0158	2.4877	-0.0019	-0.7879
only_volatility	0.0149	2.0658	-0.0027	-0.5984
drop_volatility	0.0180	3.0810	0.0004	0.4624
only_liquidity	0.0153	2.2030	-0.0024	-0.5477
drop_liquidity	0.0185	3.1947	0.0008	0.8715

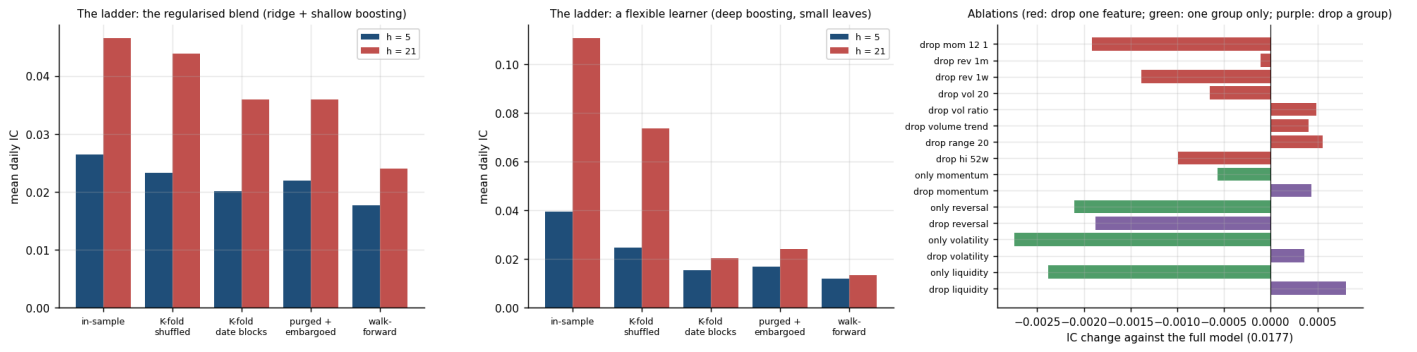


Figure 2. Left: the same learner under five validation schemes. Right: IC change of each ablation against the full model.

Neutralisation

signal	ic	lo	hi	t
raw	0.0177	0.0058	0.0299	3.0697
sector-neutral	0.0123	0.0038	0.0206	2.9823
beta, size and sector neutral	0.0086	0.0024	0.0147	2.7845
plus momentum and volatility n	0.0085	0.0027	0.0139	2.9871

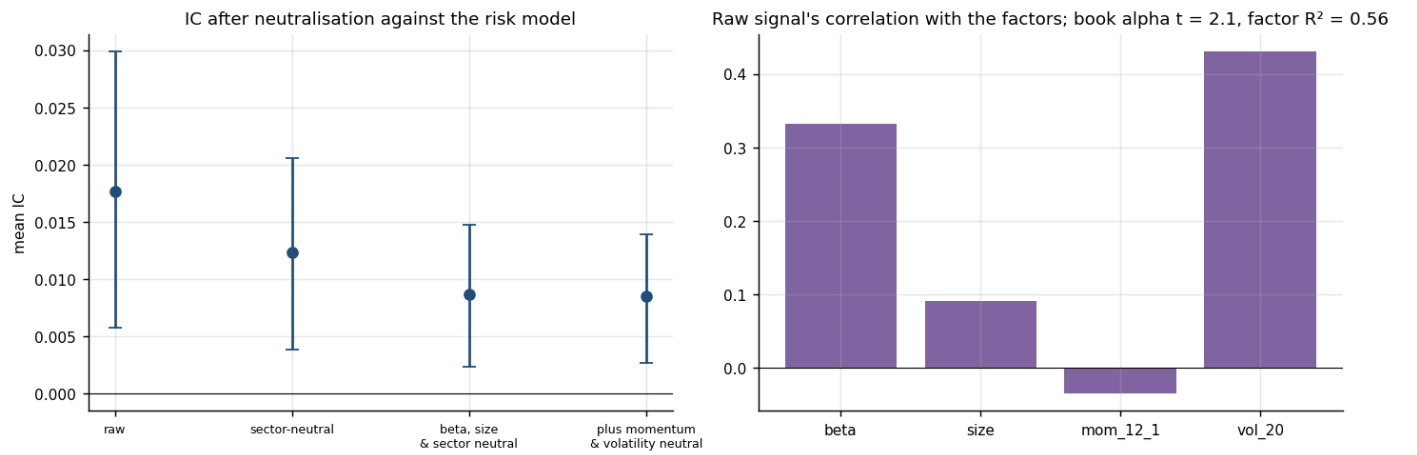


Figure 3. IC after each neutralisation, and the raw signal's correlation with the risk model's exposures.

Costs, capacity and horizon

cost model	gross_ann	net_ann	cost_ann	sharpe_gross	sharpe_net	turnover	avg_cost_bp
routed	0.0596	0.0265	0.0331	0.7378	0.3255	0.4745	6.7377
observed	0.0596	0.0167	0.0430	0.7378	0.2030	0.4745	8.6970

horizon	gross_ann	net_ann	sharpe_gross	sharpe_net	turnover	avg_cost_bp
1	0.0621	0.0039	0.7514	0.0471	0.2171	5.1679
2	0.0658	0.0169	0.8189	0.2116	0.3206	5.9008
5	0.0596	0.0265	0.7378	0.3255	0.4745	6.7377
10	0.0515	0.0356	0.7010	0.4838	0.4454	6.7793
21	0.0529	0.0466	0.6446	0.5681	0.3876	6.4391

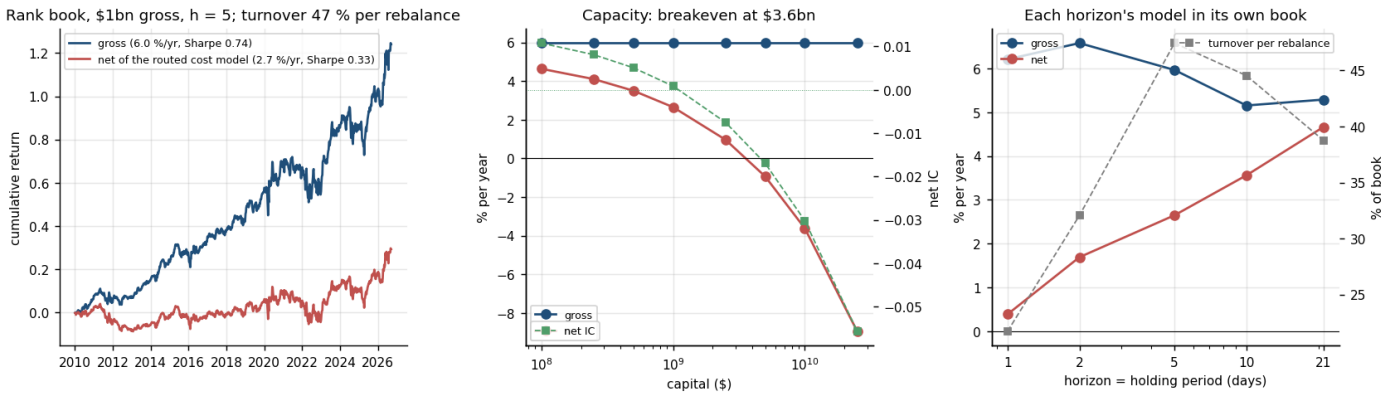


Figure 4. The rank book gross and net at \$1bn; the capacity curve; each horizon's model in its own book with its turnover.

New products

universe / model	n	ic	lo	hi	t	net_ann	sharpe_net
equity, own model	156.0000	0.0177	0.0058	0.0299	3.0697	0.0265	0.3255
etf, equity model	92.0000	0.0218	0.0054	0.0387	2.6953	-0.0315	-0.3922
etf, own model	92.0000	0.0256	0.0108	0.0409	3.4423	-0.0486	-0.6941
futures, equity model	34.0000	0.0238	0.0073	0.0392	3.1453	0.0628	0.6116
futures, own model	34.0000	0.0098	-0.0044	0.0254	1.3645	0.0340	0.3419

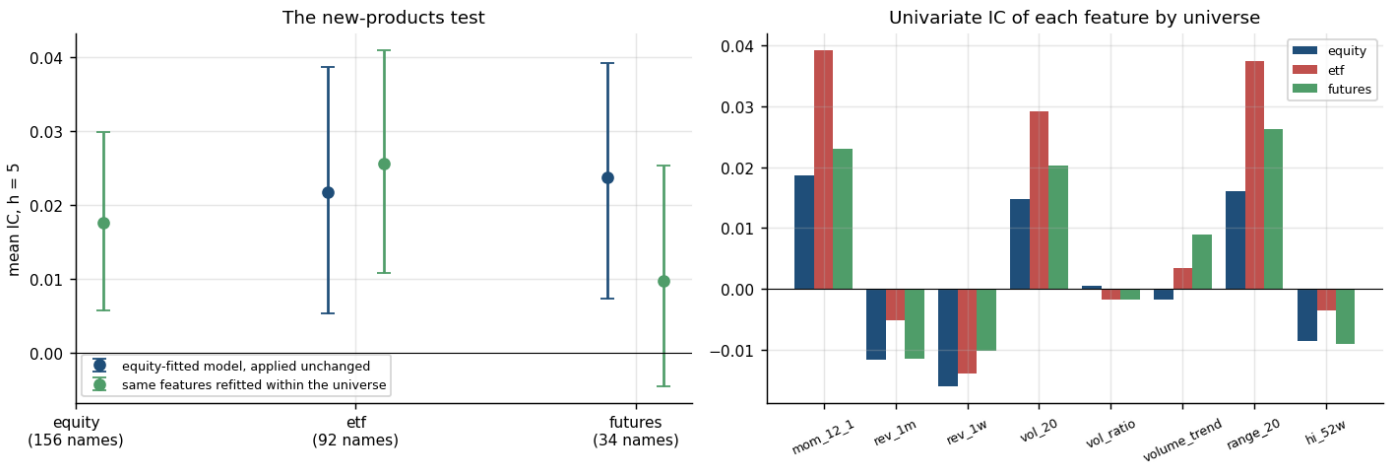


Figure 5. The equity-fitted model applied unchanged and refitted within each universe; univariate IC of each feature by universe.

What is real and what is simulated

Real: daily bars from the Yahoo chart API for 156 US large caps (the algo-wheel panel), 92 ETFs and 34 CME/ICE continuous futures; the features, targets, models and every statistic. Modelled: the cost of trading, from the algo-wheel's pre-trade model, which was itself fitted on a simulated market at the published scale of Almgren et al. (2005); the futures cost is half a tick plus a commission because the recorded contract volumes are not reliable enough for an impact term. The equity universe is today's large caps, so the gross numbers carry survivorship bias; futures are unadjusted front-month series, so roll gaps sit in their returns.

Method summary. Features rank-normalised to $N(0,1)$ per day; target the h-day forward return demeaned across the eligible universe; ridge (alpha 10) and gradient boosting (150 trees, depth 3) z-scored per day and averaged; refits each January on all earlier data with an embargo of $h + 5$ days. IC = daily Spearman correlation; t from a Bartlett HAC variance at lag h; bands from a moving-block bootstrap with blocks of $2h + 1$ days. Purged K-fold drops training dates within $h + 5$ days of the test block on either side. Neutralisation is the residual of a daily cross-sectional regression on the exposures; factor returns from the same regression on the forward return. The book is rank-weighted, dollar-neutral, capped at 2 %, rebalanced every h days with drift, and each rebalance pays the cost model per name at the order's share of ADV. Deflated Sharpe after Bailey and Lopez de Prado (2014).